MIL-C-8678 (Aer)
19 January 1954
Superseding
E-59d
20 September 1946

MILITARY SPECIFICATION

Cooling Requirements of Power Plant Installations

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 This specification covers the allowable engine installation temperatures and other applicable values, (pressure drop, heat exchanger effectiveness, etc.) that determine satisfactory cooling of an aircraft or airship power plant installation.

1.1.1 This specification covers piloted aircraft, target drone and guided missile power plant installations.

1.1.2 Section 3.2.1 covers all air cooled reciprocating engine installations.

1.1.3 Section 3.2.2 covers all turbo-jet and turbo-prop engine installations.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 GENERAL: The following specifications, publications, and drawings of the issue in effect on the date of invitation for bids, form a part of this specification.

2.2 SPECIFICATIONS:

Military Specifications

MIL-G-6099	"Generators and Regulators; Aircraft, Alternating Current, General Specification for"
MIL-G-6162	"Generators, 30 volt, Direct Current, Aircraft Engine Driven, General Specification for"

Bureau of Aeronautics Specification

SD-24	"General Specification for Design and Construction of Airplanes for the United States Navy"
SD-2000	"General Specification for Design and Construction of Pilotless Aircraft for the United States Navy"

2.3 OTHER PUBLICATIONS:

Air Force - Navy Aeronautical Bulletins

No. 421	"Atmospheric Properties - Extreme Cold and Hot: Standard for Aeronautical Design"

2.4 DRAWINGS:

Bureau of Aeronautics Drawing No. 49A1B15

2.5 AVAILABILITY OF SPECIFICATIONS, PUBLICATIONS, AND DRAWINGS:

2.5.1 When requesting specifications, publications, and drawings, refer to both title and number.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T10:46Z
Check the source to verify that this is the current version before use.

2.5.2 Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention Technical Records Division.

3. REQUIREMENTS

3.1 General Requirements

3.1.1 The Power Plant installation shall provide allowable operating temperatures and other allowable operating values as outlined in Tables I and V, and elsewhere herein at all allowable engine and airplane operating conditions as specified herein with free air temperatures on the ground and at altitude as outlined in Figure I, Curve "A", except that under Test No. 10 of Table II, and Test No. 6 of Table III, the air temperature shall be defined by Figure I, Curve "B".

3.1.2 Fixed Wing and Rotary Wing Aircraft and Airship Loadings - The power plant installation shall cool satisfactorily under all conditions of flight, as specified herein, at the maximum aircraft or airship useful load as specified in the applicable detail design specifications based on Specification SR-24.

3.1.2.1 Exceptions to Paragraph 3.1.2 shall be as follows:

1. High speed level flight at Military and Normal rated power or at Maximum and Normal rated thrust shall be made with the primary mission useful load as specified in the applicable detail design specification.

2. For reciprocating engine installations where the maximum useful load provides a service ceiling lower than the highest engine critical altitude, the power plant installation shall also provide satisfactory cooling, as specified herein, at that gross weight which will provide a service ceiling not more than 4000 feet above the highest engine critical altitude.

3. The maximum range cruise power test shall be made with the aircraft or airship in minimum drag condition less all cargo, passengers, or expendable armament and with 25% of internal fuel stores.

3.1.3 Combination Power Plant Aircraft - In any aircraft which employs a booster power plant for take-off and/or emergency operation, the main power plant(s) shall cool satisfactorily with the booster power plant(s) not operating.

3.2 Detail Requirements

3.2.1 Reciprocating Engines

3.2.1.1 Table I outlines allowable engine operating temperatures which constitute satisfactory cooling of a reciprocating engine installation under the free air temperature of Figure I, Curve "A"

3.2.1.2 Power Conditions

3.2.1.2.1 Military Rated Power - Military RPM and the manifold absolute pressure (MAP) as necessary to obtain the power(s) set forth in the applicable engine model specifications. If the required manifold pressure on a hot day exceeds the MAP limits of the engine, the MAP limits shall be used. If the MAP limits on the engine have not been determined, the MAP required to give military power on a standard day shall be used. If the engine is equipped with an automatic manifold pressure regulator, or power control, the applicable portions of the test shall be conducted in the control position designated for military power.

3.2.1.2.2 Normal Rated Power - Normal rated RPM and the manifold pressure as necessary for the power(s) set forth in the applicable engine model specification except that if the engine is equipped with automatic manifold pressure regulator or power control with a pre-established position for normal rated power, applicable portions of the test shall be conducted with the control in the designated position.

Source: https://assist.dla.mil -- Downloaded: 20260-17T10:46Z
Check the source to verify that this is the current version before use.

3.2.1.2.3 Maximum Range Power - The manifold pressure(s) and RPM determined to be the optimum settings for obtaining maximum range from the aircraft at the specified altitude and loading condition.

3.2.1.3 Reverse Pitch Propellers for Aerodynamic Braking and Ground Maneuvering - In any aircraft or airship wherein reverse pitch propeller(s) are used for the purpose of aerodynamic braking or ground maneuvering, the power plant(s) shall cool satisfactorily with the propeller(s) in the reverse pitch position during all maneuvers in which aerodynamic braking or ground maneuvering by the propellers is considered normal procedure as outlined in the applicable airplane or airship detail design specification.

3.2.1.3.1 The maximum permissible engine operating temperatures as specified in the applicable engine model specification released by the Bureau of Aeronautics shall apply when the engine-propeller combination is being used as an aerodynamic braking unit.

3.2.1.4 Intercooler Effectiveness - For installations having external or auxiliary stage supercharging, satisfactory cooling of the engine charge air shall be based on intercooler installation effectiveness. The intercooler installation effectiveness in the military power climb, at all altitudes, at the airspeed for best rate of climb shall be not less than that required to cool the engine charge air, under the ambient temperature conditions outlined in Figure 1, Curve "A", to a temperature not exceeding the carburetor air temperature corresponding to that on which engine ratings are based for auxiliary or external supercharger operation as specified in the applicable engine model specification released by the Bureau of Aeronautics.

3.2.1.4.1 Exceptions to Paragraph 3.2.1.4 shall be as follows:

1. For installations where the sea level conditions are critical and where auxiliary supercharging is not required to meet sea level performance guarantees, the critical design condition at altitude shall be used in lieu of the sea level condition.

2. The maximum intercooler installation effectiveness at the highest critical altitude in the military power climb need not exceed that determined by the following formula:

Maximum required intercooler installation effectiveness -

$$E = .80 - .0075\ h$$

Where h = highest critical altitude in thousands of feet.

Note: Intercooler Installation Effectiveness is defined as follows:

$$E = \frac{T_1 - T_2}{T_1 - T_3}$$

Where T_1 = External or Auxiliary Stage Discharge Temperature - °C.

T_2 = Carburetor Air Temperature - °C.

T_3 = Free Air Temperature - °C.

3.2.1.5 Induction System Temperature Rise

3.2.1.5.1 Alternate Air System - In airplanes equipped with an alternate air system, the temperature rise in the induction system before the carburetor or auxiliary stage supercharger, under all allowable conditions of flight and engine operation, and with the control in the "direct air" position, shall not exceed 5°C (9°F). The actual temperature rise shall be corrected for adiabatic temperature rise due to ram. If the entrance to the induction system is located on the downstream side of a cooling fan, the actual temperature rise shall also be corrected for the temperature rise through the fan.

3.2.1.5.2 Preheat System - In airplanes equipped with a carburetor air pre-heat system, the temperature rise in the induction system before the carburetor or auxiliary stage supercharger, under all allowable conditions of flight and engine operation, and with the control in the "full cold" position shall not exceed 10°C (18°F). The actual temperature rise shall be corrected for adiabatic temperature rise due to ram.

Source: https://assist.dla.mil -- Downloaded: 202640-17T10:46Z
Check the source to verify that this is the current version before use.

3.2.1.6 **Induction System Variable Alternate Air and Preheat Temperature Rise** - With the carburetor air control in any position from "Alternate" or "Hot" to "Direct" or "Cold" the heated or protected air shall be thoroughly mixed to preclude stratification and there shall be no deleterious effects on fuel metering.

3.2.1.6.1 Variable Alternate Air - With the control in the full "Alternate" position, the temperature rise over free air temperature shall be above the minimum specified in Figure 2 under the conditions of tests 4 through 10 of Table II, 1 through 8 of Table III, and 3 through 13 of Table IV, and shall be below the maximum limit specified in Figure 2 under the conditions of test No. 10 of Table II and test No. 8 of Table III.

3.2.1.6.2 Preheat - It shall be possible to obtain a temperature rise in excess of the minimum specified in Figure 2 for test 4 through 10 of Table II, 1 through 8 of Table III, and 3 through 13 of Table IV.

3.2.1.7 **Induction System Two Position Alternate Air Temperature Rise** - With the control in the "alternate" position, the temperature rise over free air temperature shall be above the minimum and below the maximum specified in Figure 2 under all test conditions.

3.2.1.8 **Generator Cooling** - Where blast cooling is required to cool reciprocating engine-driven generators, the pressure drop across the generator in flight shall be determined under the following conditions:

a. Airplanes - airspeed for maximum rate of climb at sea level and at service ceiling.

b. Helicopters - zero airspeed at sea level and at hovering ceiling.

c. Airships - zero airspeed at sea level and at the pressure height for 80% helium fullness at take-off.

The ΔP_{req}, referred to herein, has no physical meaning and is used solely for the purpose of determining that the installation is satisfactory for cooling any generator of a like rating which is approved under the same specifications.

3.2.1.8.1 On generators, conforming to Paragraphs 3.4.1.9 and 3.3.11 of Specifications MIL-G-6099 and MIL-G-6162 respectively, the generator cooling air duct system shall provide a pressure drop across the generator equal to:

$$\Delta P_{req} = (P_{T_1} - P_{S_3}) - \left[\frac{(P_{T_1} - P_{S_3})^{1.88} - \Delta P_X^{1.88} \left(\frac{X}{100}\right)^2}{1.89}\right] \left(\frac{\quad}{\quad}\right)^2$$

Where:

ΔP_{req} = Pressure drop required across the generator in "H_2O

P_{T_1} = Blast tube inlet total pressure

P_{S_3} = Cooling system exit static pressure

ΔP_X = Pressure drop in "H_2O required across the generator to obtain 100 percent generator rating at all altitudes. (Obtain this value from BuAer Drawing No. 4[illegible] or the applicable generator detail drawing using the inlet total temperature which would occur in flight with ambient temperatures in accordance with Figure 1, Curve A.)

ΔP_m = Measured pressure drop across the generator in "H_2O (i.e. $P_{T_2} - P_{S_3}$, where P_{T_2} = Total pressure at the entrance to the generator.)

X = % normal rated cooling airflow at which the generator is approved in accordance with Paragraph 3.3.11 and Figure 3 of Specification MIL-G-6162 or in accordance with Paragraph 3.4.1.9 and Figure 2 of Specification MIL-G-6099 and the applicable generator detail drawing.

Source: https://assist.dla.mil -- Downloaded: 202640-17T10:46Z
Check the source to verify that this is the current version before use.

Note: Where $(P_{T_1} - P_{S_3})$ is less than 7" H_2O the required pressure drop across the generator shall be:

$$\Delta \text{ Preq} = 7 - \left(\frac{X}{100}\right)^2$$

3.2.1.8.2 On generators which deviate from paragraphs 3.4.1.9 and 3.3.11 of Specifications MIL-G-6099 and MIL-G-6162 respectively, the generator cooling air duct system shall provide a pressure drop across the generator equal to:

$$\Delta \text{ Preq} = (P_{T_1} - P_{S_3}) - \left[\frac{(P_{T_1} - P_{S_3}) - .77\ \Delta Px}{.76}\right] \left[\frac{\Delta Pm \left(\frac{X}{100}\right)^2}{16 - \left(\frac{X}{100}\right)^2}\right]$$

Note: Nomenclature same as in Section 3.2.1.8.1, except as follows:

Where $(P_{T_1} - P_{S_3})$ is less than 16" H_2O, the required pressure drop across the generator shall be:

$$\Delta \text{ Preq} = 16 - \left(\frac{X}{100}\right)^2$$

3.2.2 Turbo-jet and Turbo-prop Engines.

3.2.2.1 The temperatures of engine components and accessories shall not exceed the limits established by the engine manufacturer, and those given in Table V, under the conditions of Figure 1, Curve "A".

3.2.2.2 Thrust and Power Conditions.

3.2.2.2.1 Maximum Thrust or Power - Thrust or Power obtained with the power control lever against the maximum thrust stop.

3.2.2.2.2 Military Rated Thrust or Power - Thrust or Power obtained with the power control lever in the military rated thrust position on the pilot's control quadrant.

3.2.2.2.3 Normal Rated Thrust or Power - Thrust or Power obtained with the power control lever in the normal rated thrust position on the pilot's control quadrant.

3.2.2.3 Reverse Pitch Propellers or Reverse Engine Thrust for Aerodynamic Braking and Ground Maneuvering - In any aircraft wherein reverse pitch propeller(s) or reverse engine thrust are used for the purpose of aerodynamic braking or ground maneuvering, the power plant(s) shall cool satisfactorily with the propeller(s) in the reverse pitch position or the engines in the reverse thrust position during all maneuvers in which aerodynamic braking or ground maneuvering by the propellers or engines is considered normal procedure as outlined in the applicable airplane detail design specification.

3.2.2.3.1 The maximum permissible engine operating temperatures as specified in the applicable engine model specification released by the Bureau of Aeronautics shall apply when the engine or engine propeller combination is being used for aerodynamic braking or for ground maneuvering.

3.2.2.4 Generator Cooling - Where blast cooling is required to cool turbine-engine driven generators, the pressure drop across the generator in flight shall be determined under the following conditions:

a. Airplanes - airspeed for maximum endurance at highest practical altitude.

b. Helicopters - zero airspeed at sea level and hovering ceiling.

3.2.2.4.1 The requirements for generator cooling on turbo-jet and turbo-prop engines are the same as those given in Section 3.2.1.8.1 and 3.2.1.8.2.

Source: https://assist.dla.mil -- Downloaded: 202640-17T10:46Z
Check the source to verify that this is the currentversion before use.

4. TEST PROCEDURES

4.1 Reciprocating Engines.

4.1.1 Table II, herein, outlines the airplane operating conditions which determine satisfactory cooling of a reciprocating engine power plant installation.

4.1.2 Table III, herein, outlines the airship operating conditions which determine satisfactory cooling of a reciprocating engine power plant installation.

4.1.3 Table IV, herein, outlines the helicopter operating conditions which determine satisfactory cooling of a reciprocating engine power plant installation.

4.2 Turbo-Jet and Turbo-Prop Engines.

4.2.1 Table VI, herein, outlines the airplane operating conditions which determine satisfactory cooling of a turbo-jet or turbo-prop power plant installation.

4.2.2 Table VII, herein, outlines the helicopter operating conditions which determine satisfactory cooling of a turbine engine installation.

Source: https://assist.dla.mil -- Downloaded: 202610-17T10:46Z
Check the source to verify that this is the currentversion before use.

TABLE I

Reciprocating Engine Temperature Limits

Item Number	Item	Correction Factor	Temp. Limits °C	Notes
1	Cylinder Head (Bayonet type thermocouple)	1.0	*	
2	Cylinder Base (Imbedded thermocouple)	.8	*	
3	Oil-In	1.0	85	100°C on the ground
4	Temp Rise Fuel - tank to carburetor	None	5	
5	Engine Flexible Mount	1.0	#	
6	Magneto	.7	85	Coil core temp.
7	Ignition Generator	.7	85	Coil core temp. (Low Tension Ignition System)
8	Ignition Step-Up Coil	1.0	85	Coil Core Temp. (Low Tension Ignition System)
9	Accessory Compartment	1.0	70	
10	Spark Plug Elbow	1.0	#	
11	Exhaust Gas to Turbo Nozzle Box	None	#	
12	Exhaust Shroud	None	350°	

NOTE: * The corrected hottest head or base temperature shall not exceed that specified in the applicable engine specification released by the Bureau of Aeronautics. For ground cooling tests, the cylinder head and base temperature limits as specified in the applicable engine model specification for take-off power shall apply unless cylinder temperature limits specifically applicable to ground operation are listed. For maximum range cruise power operation, the cylinder head temperature of the coldest cylinder shall not be less than that specified by the engine manufacturer. If no minimum temperature limit is specified by the engine manufacturer, a minimum limit of 100°C shall apply. For such operation, alternate air may be used if necessary to maintain engine stability.

The corrected temperature shall not exceed that specified in the applicable engine, turbo supercharger or engine accessory specification released by the Bureau of Aeronautics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:46Z
Check the source to verify that this is the current version before use.

TABLE II

Ground and Flight Conditions on Which Reciprocating Engine Airplane Cooling Requirements are Based

Test No.	Condition	Altitude	Power	RPM	Mixture	Cowl Flaps or Shutters	Applicable Note
1	Ground	0	-----	60%	Rich	Open	1,2,3,12
2	Water	0	-----	60%	Rich	Open	2,4
3	Ground	0	-----	50%	Rich	Open	2,5
4	Ground	0	-----	60%	Rich	Open	1,2,6
5	Climb	---	Military Rated	Military Rated	Normal	Open	7
6	Level	Highest Critical	Military Rated	Military Rated	Normal	As Required	10,12,14
7	Level	Highest Critical	Normal Rated	Normal Rated	Normal	As Required	9,10,12
8	Level	Highest Critical	65% Normal Rated	Minimum Allowable	Normal	As Required	8,10,13
9	Level	2,000 ft.	Military Rated	Military Rated	Normal	As Required	10,14
10	Level	3,000 ft.	Maximum Range Power	---	Normal	Closed	11,13

NOTES:

1. Ground runs shall be conducted with the propeller in take-off adjustment and with the airplane pointed cross wind but with as little wind as possible, and in no case, with a wind velocity in excess of 10 mph.

2. The rpm shall be the designated percent of maximum rpm and shall be obtained with the propeller in take-off adjustment. The idling speed shall be the minimum rpm for satisfactory operation.

3. Run for 10 minutes, idle for 5 minutes, and shut down measuring shut down temperatures until they have peaked.

4. Applicable only to sea planes and amphibians. This test shall consist of taxiing both up wind and down wind with a 12-18 knot true wind velocity at 60% rpm (note 2). Also, taxiing tests shall be conducted with a 12-18 knot true wind velocity on the port beam, with minimum practicable rpm on the starboard engine(s), and with the necessary rpm on the port engine(s) to maintain a straight course. Repeat with the wind on the starboard beam. Each run shall be of not less than 5 minutes duration and until temperatures stabilize.

5. Applicable only to airplanes designed for carrier based operation. The test shall be conducted with a true wind velocity of 12-18 knots first on the port beam and then on the starboard beam.

6. Run for 10 minutes, idle for 5 minutes, and follow immediately with test No. 5. Take-off may be accomplished with "Rich" carburetor setting shifting to "Normal" immediately after take-off.

7. Climb shall be conducted at the airspeed of maximum rate of climb and shall be continued to the service ceiling of the airplane. If engine military power time limit is reached below the service ceiling, continue the climb at normal rated power.

8. This cruise test shall be conducted with the maximum drag condition of contemplated external tanks, bombs, torpedoes, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:46Z
Check the source to verify that this is the current version before use.

9. This requirement applies only where normal rated power critical altitude is at least 5000 feet higher than military rated power critical altitude.

10. Record cowl flap and shutter settings required to provide cooling under the ambient conditions shown on Curve "A", Figure 1.

11. If auxiliary cooling controls are provided for use during extreme cold weather, they shall be set in the minimum cooling position.

12. Tests shall be conducted with the carburetor air control in the "Direct" or "Cold" position and repeated with the control positioned to obtain the minimum allowable carburetor air temperature or above.

13. Tests shall be conducted with the carburetor air control in the "Direct" or "Cold" position and repeated with the control in the full "Hot" or "Alternate" position.

14. Tests shall be continued until temperatures stabilize or until the military power time limit is reached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:46Z
Check the source to verify that this is the current version before use.

TABLE III

Ground and Flight Conditions on Which Reciprocating Engine Airship Cooling Requirements are Based

Test No.	Condition	Pressure Altitude	Power	RPM	Mixture	Cowl Flaps or Shutters	Applicable Notes
1	Ground	0	-	50%	Rich	Open	1,9
2	Climb	-	Military Rated	Military Rated	Normal	Open	2,11
3	Level	Highest Critical	Military Rated	Military Rated	Normal	As Required	8,9,11,12,13
4	Level	Highest Critical	65% Normal Rated	Minimum Allowable	Normal	As Required	3,8,10,11,12
5	Level	500 ft.	Military Rated	Military Rated	Normal	As Required	4,8,11
6	Hover	500 ft.	As Required	As Required	Normal	As Required	5,8,11
7	Turns	500 ft.	Military Rated	Military Rated	Normal	As Required	6,8,11
8	Level	3000 ft.	Maximum Range	-	Normal	Closed	7,10,11

NOTES:

1. Ground runs shall be conducted with the propeller in take-off adjustment and with the airship pointed upwind with a 12-18 knot true wind velocity.
2. Climb shall be conducted at the airspeed for maximum rate of climb and shall be continued to the airship pressure height for 80% helium fullness at take-off. If engine military power time limit is reached below the above pressure height, continue climb at normal rated power.
3. This cruise test shall be conducted with the maximum contemplated drag condition of bombs, torpedoes, etc.
4. This test shall be continued until temperatures stabilize and followed immediately by test No. 6. On ASW airships, this test shall be conducted using the necessary manifold pressure and RPM such that the maximum design towing speed is obtained while exerting the maximum design towing thrust.
5. This test shall be conducted by decelerating from the speed of test No. 5 using reverse pitch, and then hover for five additional minutes.
6. This test shall be conducted by doing eight (8) consecutive 360° turns with full rudder deflection, measuring temperatures on the engine on the side of the direction of the turn.
7. If auxiliary cooling controls are provided for use during extreme cold weather, they shall be set in the minimum cooling positions.
8. Record cowl flap and shutter settings required to provide cooling under the ambient conditions shown on Curve "A", Figure 1.
9. Tests shall be conducted with the carburetor air control in the "Direct" or "Cold" position and repeated with the control positioned to obtain the minimum allowable carburetor air temperature or above.
10. Tests shall be conducted with the carburetor air control in the "Direct" or "Cold" position and repeated with the control in the full "Alternate" or "Hot" position.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T10:46Z
Check the source to verify that this is the current version before use.

11. These tests will be conducted with the airship loaded to 2000 pounds static heaviness.

12. These tests shall be conducted at the designated altitude or at the pressure height for 80% helium fullness at take-off, whichever is lower.

13. This test shall be continued until temperatures stabilize or until the military power time limit is reached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:46Z
Check the source to verify that this is the current version before use.

Ground and Flight Conditions on Which Reciprocating Engine Helicopter Cooling Requirements are Based

Test No.	Condition	Pressure Altitude	Power	RPM	Mixture	Applicable Notes
1	Ground (Rotor Engaged)	0	-	60%	Rich	1,7,8
2	Ground (Rotor Engaged)	0	-	60%	Rich	2,8
3	Ground (Rotor Engaged)	0	-	60%	Rich	3,8
4	Climb (Forward Flight)	-	Military Rated	Military Rated	Normal	4,6
5	Climb (Vertical)	-	Military Rated	Military Rated	Normal	5,6
6	Level Forward Flight	Service Ceiling	Military Rated	Military Rated	Normal	6,7
7	Level Forward Flight	Highest Critical	Normal Rated	Normal Rated	Normal	7
8	Level Forward Flight	500 ft.	Normal Rated	Normal Rated	Normal	7
9	Level Forward Flight	Service Ceiling for 70% NRP	70% NRP	Minimum Allowable	Normal	7
10	Level Forward Flight	500 ft.	70% NRP	Minimum Allowable	Normal	7
11	Hover	Out of ground Effect	-	Minimum Allowable	Normal	7
12	Hover	Out of ground effect	-	Normal Rated	Normal	7
13	Hover	Hovering Ceiling	Normal Rated	Normal Rated	Normal	7

NOTE:
1. This test shall be run with the helicopter pointed first upwind, then downwind, then with wind on either beam. Tests shall be run with a 12-18 knot true wind velocity.
2. Run for 10 minutes, idle for 5 minutes and shut down measuring shut down temperatures. The test shall be run with the helicopter pointed cross wind, but with as little wind as possible.
3. Run for 10 minutes, idle for 5 minutes, and follow immediately with Test No. 4.
4. Climb shall be conducted at airspeed for best rate of climb and shall be continued to the service ceiling. If engine military power time limit is reached below the service ceiling, continue climb at normal rated power.
5. Climb shall be continued to hovering ceiling at zero forward airspeed.
6. Normal rated power shall apply if no military rating is assigned.

Source: https://assist.dla.mil -- Downloaded: 202610-17T10:46Z
Check the source to verify that this is the current version before use.

7. These tests shall be continued until temperatures stabilize.

8. RPM for these tests shall be the designated percent of the military rated RPM.

Source: https://assist.dla.mil -- Downloaded: 202610-17T10:46Z
Check the source to verify that this is the current version before use.

GROUP 11

TABLE V

Turbo-Prop and Turbo-Jet Engine Temperature Limits

Item No.	Item	Correction Factor	Temperature Limit
1	Exhaust Shroud	*	**
2	Structure adjacent to tailpipe or afterburner (at least 6 points)	*	***
3	All applicable accessories	1.0	****
4	Oil-in	1.0	*****

NOTE:

* No correction factor shall be used for these items.

** Use limit established by the engine or airframe manufacturer.

*** Use limit established by the airframe manufacturer.

**** Use limit established by the accessory manufacturer.

***** Use limit established by the engine manufacturer. Item 4 applicable only if oil cooler is furnished by the airframe manufacturer.

Source: https://assist.dla.mil -- Downloaded: 202610-17T10:46Z
Check the source to verify that this is the current version before use.

TABLE VI

Ground and Flight Conditions on Which Turbine Engine Airplane Cooling Requirements are Based

Ground Tests

Test No.	Instructions
1	Operate engine for two minutes at Normal Rated Thrust or Power, Idle for five (5) minutes, then shut down, measuring shutdown temperatures until they have peaked. This test shall be conducted with as little wind as possible and repeated with a true wind velocity of 12-18 knots first on the port beam and then on the starboard beam.
2	This test is applicable to seaplanes and amphibians only and shall consist of taxiing upwind and downwind with a 12-18 knot true wind velocity at the thrust or power necessary to maintain the airplane 1 to 5 knots under the minimum airspeed required to keep the airplane on the step or water skis. Also, in multi-engine airplanes, taxiing tests shall be conducted with a 12-18 knot true wind velocity on the port beam, with idle setting on the starboard engine(s) and the necessary thrust or power setting on the port engine(s) to maintain a straight course. Repeat with the wind on the starboard beam. Each run shall be continued until temperatures stabilize.

Flight Tests

Test No.	Instructions
* 3	Conduct a military thrust or power climb at the airspeed of best rate of climb to the airplane service ceiling. If engine military thrust or power time limit is reached below the service ceiling, continue the climb at Normal Rated Thrust or Power.
** 4	Conduct a level flight run at maximum thrust at approximately 2000 feet pressure altitude for 3 minutes (Turbo-jets only).
** 5	Conduct a level flight run at military thrust or power at approximately 2000 ft. pressure altitude until temperatures stabilize or for ten (10) minutes. (This test need not be run if test #4 has been accomplished).
6	Conduct a level flight run at the thrust or power and altitude required for maximum range until temperatures stabilize.

* When the use of maximum thrust for climb is a normal procedure as outlined in the applicable airplane detail design specification, this test shall be repeated using maximum thrust.

** These tests shall be limited to the thrust or power which will provide an airspeed not exceeding the operating or structural limits of the aircraft or engine when such limits exist.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T10:46Z
Check the source to verify that this is the current version before use.

TABLE VII

Ground and Flight Conditions on Which Turbine Engine Helicopter Cooling Requirements are Based

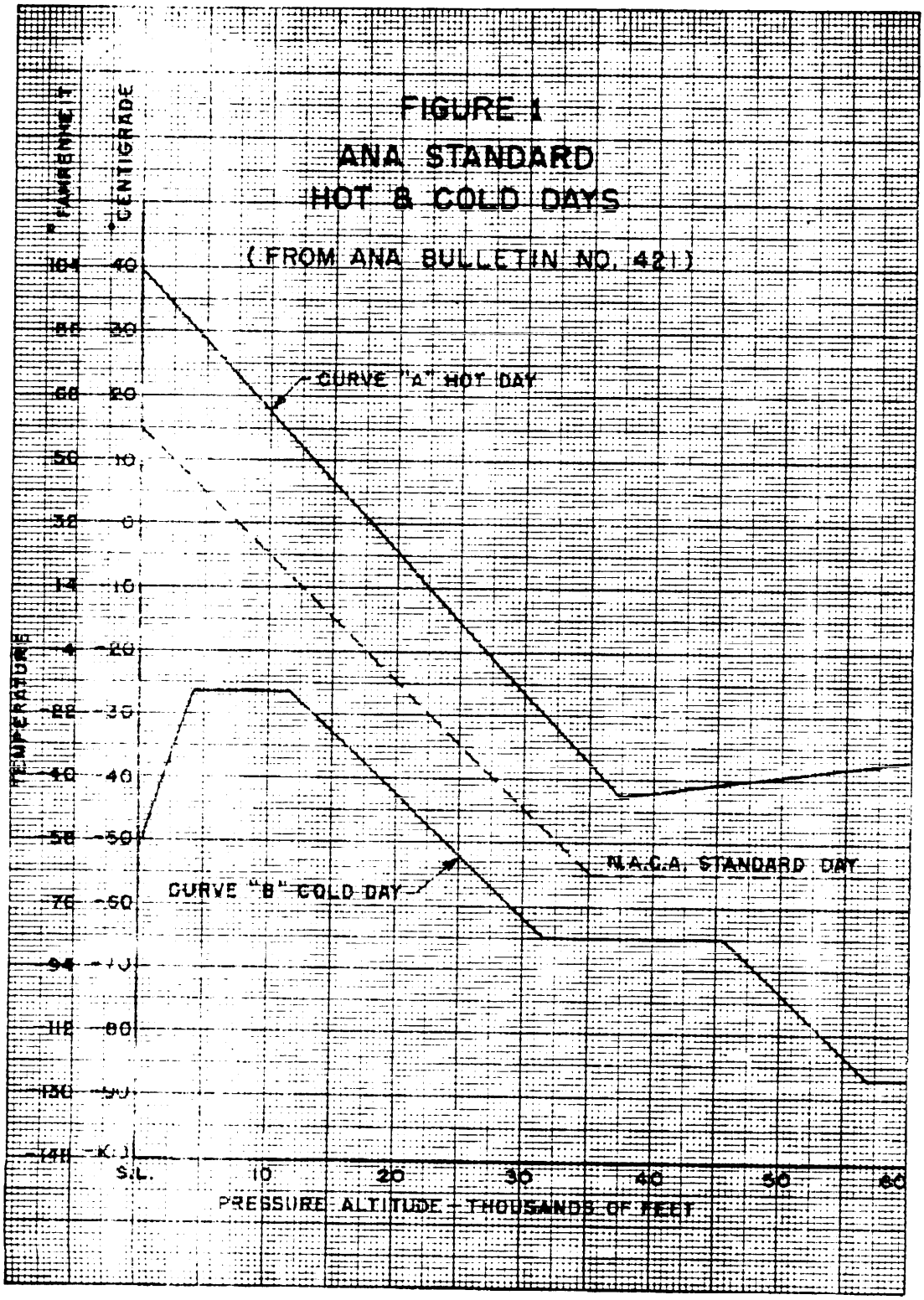
Ground Tests

Test No.	Instructions
1	Operate engine for two minutes at Normal Rated Thrust or Power. Idle for five (5) minutes, then shut down, measuring shutdown temperatures until they have peaked. This test shall be conducted with as little wind as possible and repeated with a true wind velocity of 12-18 knots first on the port beam and then on the starboard beam.

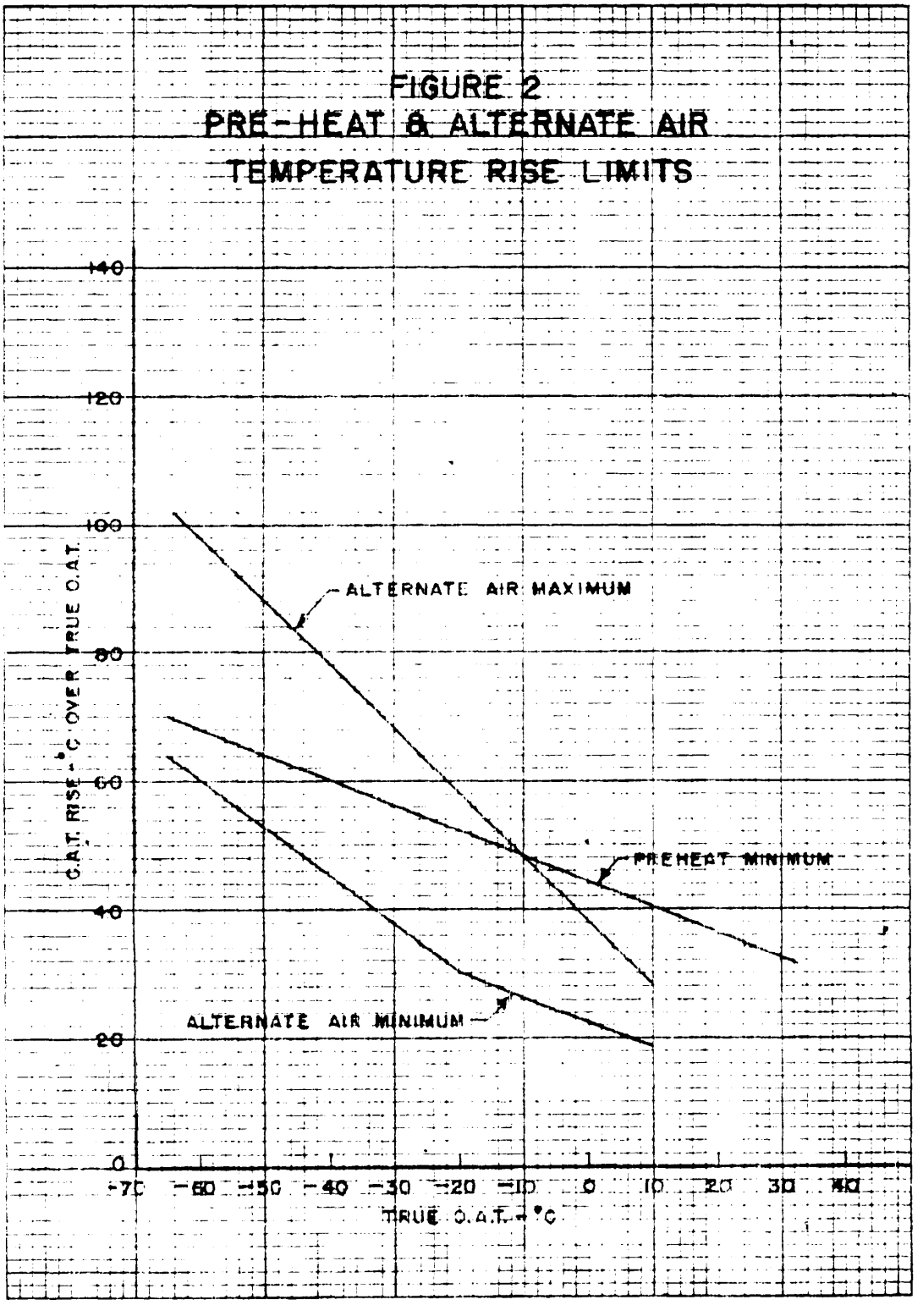
Flight Tests

Test No.	Instructions
2	Conduct a military thrust or power climb at the airspeed of best rate of climb to the helicopter service ceiling. If engine military thrust or power time limit is reached below the service ceiling, continue the climb at Normal Rated Thrust or Power.
3	Conduct a military thrust or power vertical climb to the hovering ceiling at zero forward airspeed. If engine military thrust or power time limit is reached below the hovering ceiling, continue the climb at Normal Rated Thrust or Power.
4	Conduct a level flight run at military thrust or power at approximately 2000 ft pressure altitude until temperatures stabilize or for ten (10) minutes.
5	Conduct a level flight run at the thrust or power and altitude required for maximum range until temperatures stabilize.
6	Conduct a hovering test out of ground effect at normal rated RPM until temperatures stabilize.
7	Conduct a hovering test at hovering ceiling using Normal Rated Thrust or Power until temperatures stabilize or for ten (10) minutes.

Source: https://assist.dla.mil -- Downloaded: 202640-17T10:46Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 202610-17T10:46Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 202610-17T10:46Z
Check the source to verify that this is the current version before use.